



Braude College of Engineering · Software Engineering Department
Capstone Project - Phase A (61998)

Multi-Modal LLM Style Analysis

Detecting Misattributed Creations in the Shakespeare Corpus through LLM-Based Style Normalization

Project Code: 26-2-R-6

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GitHub: github.com/eliranmeleh/Multi-Modal-LLM-Style-Analysis

The problem & the need

- **The question.** Which of the 49 creations attributed to Shakespeare are genuinely his, and which bear another hand? An ideal computational tool for this question would have three properties:

Judges a text on its own

No external pool of comparison authors to gather - and nothing to rebuild for each new author.

Needs no training

Works out of the box - no labeled examples and no models to train.

Explains itself

A result a literary scholar can read in plain stylistic terms - not just a score.

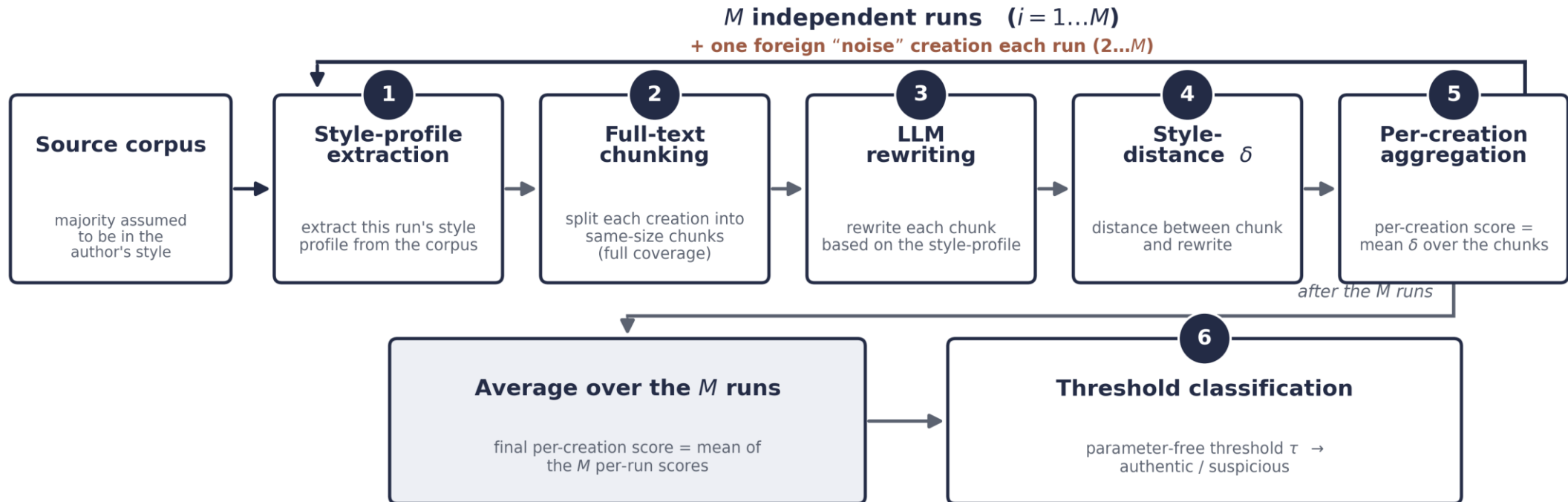
The gap: existing computational methods rarely offer all three at once - so this project sets out to build one that does: **unsupervised, interpretable, and transferable, complementing existing approaches.**

Project goals

- **Unsupervised & zero-shot.** Design and implement an authorship-anomaly pipeline driven by LLM-based style normalization - no labels, no training.
- **Interpretable.** Produce results a literary scholar can read directly - a per-creation score in plain stylistic terms, not an opaque number.
- **Methodological cross-check.** Benchmark against existing stylometry findings on the corpus as a complementary view of the same question.
- **Generalizable.** The same pipeline applies to any author's corpus by substitution - Shakespeare is only an example, with no “Shakespeare” hard-coded anywhere.
- **Open & reproducible.** Deliver an open-source implementation other researchers can reuse and extend.

Methodology - pipeline overview

The core idea - rewrite each passage into the corpus's own style - how much it must change is the distance the method measures (small change → authentic, large → another hand).

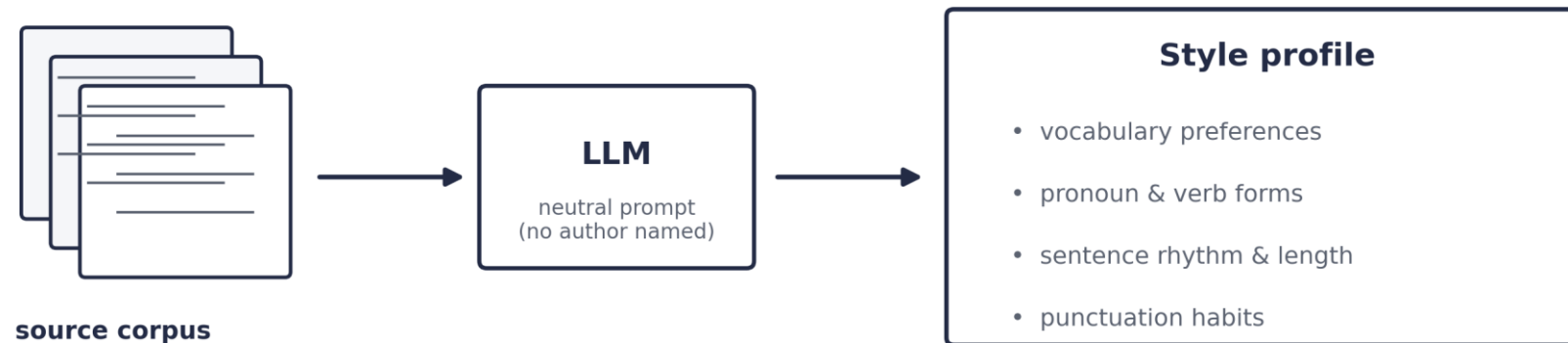


Main proposal: runs $2 \dots M$ each add one different foreign "noise" creation (run 1: none; non-cumulative).

Phase B also evaluates a no-added-noise variant (the LLM's intrinsic run-to-run variability).

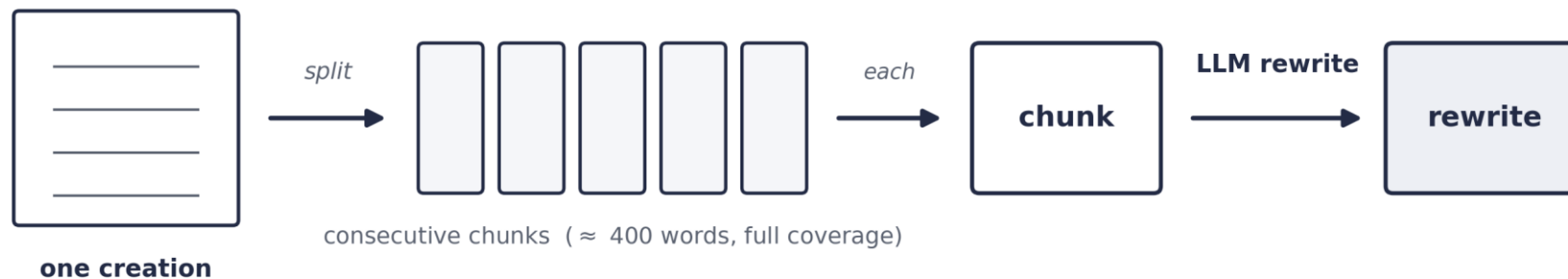
Step 1 - Style-Profile Extraction

- **Input.** The full text of every creation in the corpus (most are assumed to be already in the author's style) - whole creations, not samples.
- **What the LLM does.** Produces a generic style profile - vocabulary, pronoun and verb forms, sentence rhythm, punctuation.
- **Neutral prompt.** No mention of “Shakespeare” or any author, so the profile carries no memorized-author bias.
- **Noise injection (main proposal).** Each profiling run after the first adds one different foreign creation as “noise” to the corpus before extraction (run 1: none; non-cumulative), so no single intruder can bias the profile; Phase B also tests with no added noise.



Steps 2 & 3 - Chunking and LLM Rewriting

- **Step 2 - Chunking.** Split each creation into consecutive same-size chunks ($P \approx 400$ words each). Full coverage - every word, no sampling.
- **Step 3 - Rewriting.** The LLM rewrites each chunk into the profile's style - preserve the content, change only the style. *How much it must change is exactly the distance the method measures (Step 4).*



Preserve the content — change only stylistic features.

Step 4 - Function-Word Edit Distance (δ)

$$\delta(chunk, rewrite) = \frac{edits(FW(chunk), FW(rewrite))}{\max(|FW(chunk)|, |FW(rewrite)|)}$$

① **Authentic chunk** — already matches the corpus profile

FW(original) and thou that it is not so for the doth

FW(rewrite) and thou that it is not so for the doth

edits = 0 → $\delta = 0 / 10 = 0.00$ (authentic — nothing to change)

② **Foreign chunk** — model must alter function words to match the profile

FW(original) and **you** that it is not so for the **does**

FW(rewrite) and **thou** that it is not so for the **doth**

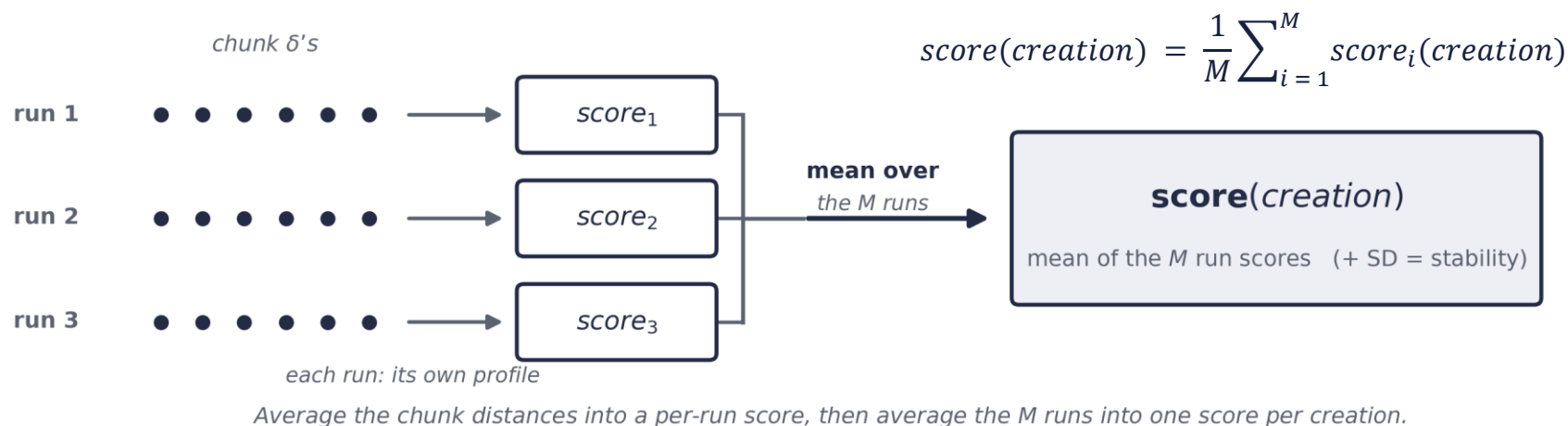
edits = 2 → $\delta = 2 / 10 = 0.20$ (suspicious — 2 of 10 function words changed)

- **FW(text).** The ordered sequence of ~120 function words - modern (you, has) and Early Modern (thou, hath) - with no author-specific cue baked in.
- **Read δ directly.** The share of function words the rewrite had to change - a chunk already in the corpus's style barely changes ($\delta \rightarrow 0$); one by another hand needs more rework (δ large).
- **One example metric.** Function-Word Edit Distance is one of several distance metrics compared in Phase B.

Step 5 - Aggregation & Robustness

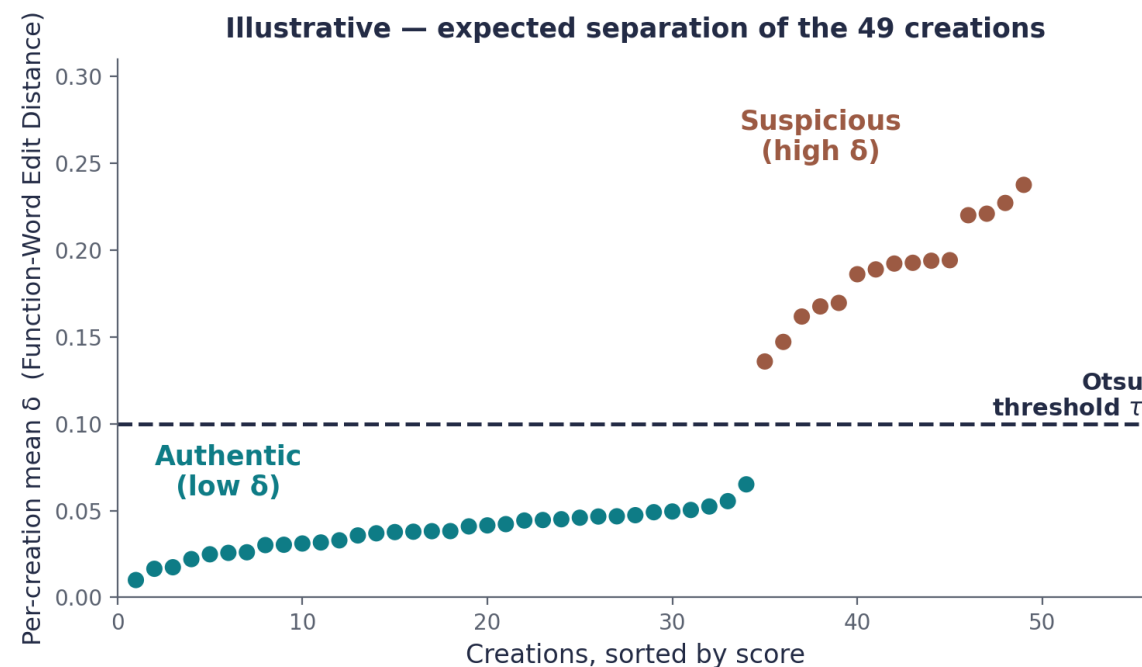
- **Aggregation.** Average δ over a creation's chunks $\rightarrow$ mean over the independent runs $\rightarrow$ one score per creation.
- **Independent runs.** The pipeline runs several times, each re-extracting its profile; averaging cancels the LLM's run-to-run randomness.

$$score_i(creation) = \frac{1}{|chunks(creation)|} \sum_{chunk \in creation} \delta(chunk, rewrite_i)$$



Step 6 - Threshold Classification

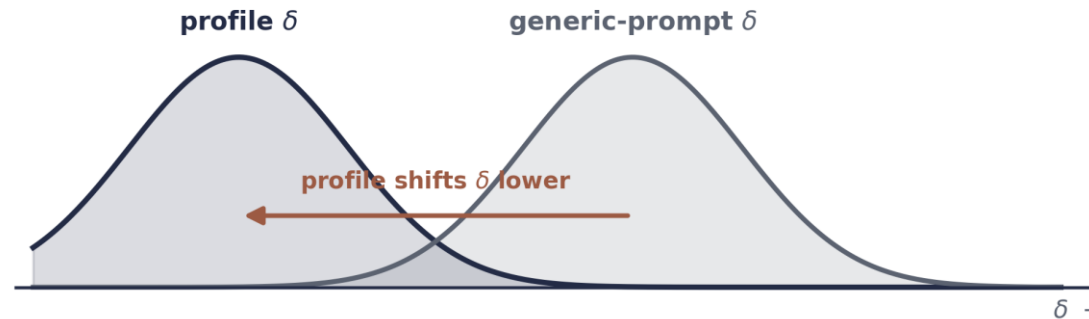
- **Threshold split.** Otsu's parameter-free threshold τ - one option illustrated here, of several classifiers (e.g. Gaussian-mixture) compared in Phase B - splits the 49 scores into authentic vs. suspicious; it fits because the suspicious class is not a rare minority.
- **Single round (Phase A).** Each run re-extracts its profile $\rightarrow$ scores averaged over the independent runs $\rightarrow$ one threshold split, with no feedback loop. Iterative purification is deferred to Phase B, adopted only if the single-round threshold proves unstable.



Evaluation - success metrics

Success metrics (checked in Phase B)

- **Profile contribution.** Profile $\delta <$ generic-prompt δ (paired t-test, $\alpha = 0.05$).
- **Agreement with scholarly consensus.** The method's suspicious set matches established scholarly-consensus co-authorship findings (e.g. Henry VIII / Fletcher, Henry VI / Marlowe) on at least 4 of 5 firmest cases.



A significant gap (paired t-test, $\alpha = 0.05$) confirms the extracted profile adds information beyond the model's prior.

Expected outputs: sorted scatter of all 49 scores with the threshold marked, a labeled suspicious-creations list with continuous scores, and per-creation confidence.

How the method will be validated

Check	Procedure	Passes if
Mixture Detection	Constructing a text composed mostly of Author A with a small injection of Author B.	The method distinguishes between them
Stability - noise (main proposal)	One different foreign creation injected per run.	Authentic labels stay stable
Calibration - runs M	Rewriting the same chunks M times.	δ stable across the M runs; otherwise raise M and repeat

Phase B - work plan & key risks

1

Reproducibility & sensitivity

Independent runs on identical inputs; vary P and check label stability.

2

Full-corpus run

All 49 creations, single round; scatter + Otsu; record the suspicious set.

3

Optional refinement

Re-extract the profile from authentic-only creations and repeat.

Constraints & risk mitigation: LLM memorization → control experiment / held-out corpus · run-to-run randomness → average several independent runs & pin model · long-context paraphrasing → short chunks & sensitivity sweep · non-bimodal scores → Gaussian-mixture fallback.

Tools & technologies

Tool	Purpose
Python 3.11	Primary implementation language
Pipeline LLM	Style extraction & rewriting; needs a long context window - the model is chosen in the POC, judged by rewrite fidelity & run-to-run stability
Gemini 2.5 / GPT-class / Claude	Candidate pipeline LLMs compared on the pilot subset
python-Levenshtein	Function-word edit-distance computation
scikit-image	Otsu's automatic 1-D threshold
scikit-learn	Comparison classifiers (cross-check vs. the prior baseline)
NumPy / Pandas	Numerical & tabular data handling
Matplotlib / Seaborn	Scatter plots, histograms, convergence curves
Project Gutenberg	Text corpus - 49 creations (identical to the prior study)
Git / GitHub	Version control, prompt/code sharing, submission artifact

Summary & Phase B outlook

- **What Phase A delivers.** A fully specified, unsupervised, interpretable, transferable LLM-based method for authorship-anomaly detection - with metric, algorithm, evaluation, and risks all defined.
- **Why it matters.** It needs no comparison corpus and no training, and its output reads directly in stylistic terms - a complementary cross-check to existing methods.

Unsupervised

Interpretable

Transferable

What's next

- Phase B implements and validates the method end-to-end, then compares the distance-metric family and reports which separates best.
- Deliver an open-source, reproducible repository with all results and plots.



Thank you

Questions & discussion

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